Experiment 8

Nmap Service Version and OS Detection

Aim

To perform service version detection and operating system identification using Nmap.

Theory

Nmap (Network Mapper) is an open-source network scanning tool used for network discovery and security auditing.

Commands

i) Detect Service Versions

nmap -sV 192.168.1.10

Description:

Determines the versions of services running on target ports.

ii) Aggressive Scan

nmap -A 192.168.1.10

Description:

Performs OS detection, version detection, script scanning, and traceroute.

iii) Operating System Detection

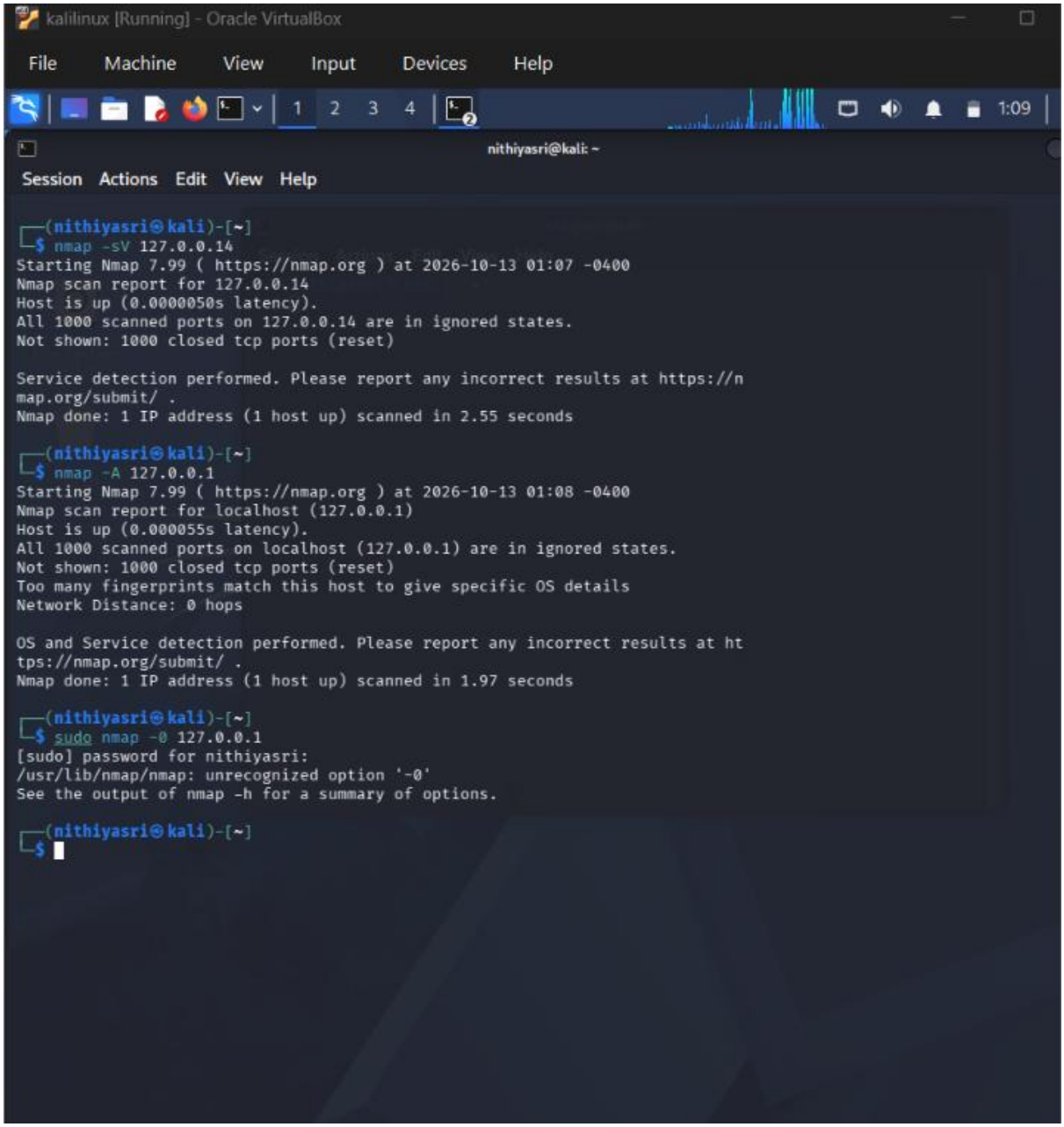
sudo nmap -O 192.168.1.10

Description:

Identifies the operating system running on the target host.

Sample Output

```
PORT     STATE SERVICE VERSION
22/tcp   open  ssh     OpenSSH 8.2
80/tcp   open  http    Apache 2.4
```



Result

Successfully performed service version detection and OS identification using Nmap.